THE CONNECTIVITY WINDOW OF BILATERAL PREFERENCE RANDOM GRAPHS

YITZCHAK SHMALO

Abstract. In the bilateral preference random graph $LK(n,k)$ of La and Kabkab, independent uniform scores are placed on the edges of K_n, each vertex prefers its k incident edges of smallest score, and an edge is kept when both of its endpoints prefer it. Dabirian and Subramanian showed that $LK(n,k)$ is connected with high probability when $k \geq \log n + t \log\log n\sqrt{\log n}$ with $t > 1/2$ and disconnected when $t < -1/2$, and left the window between these two conditions open. We show that for every fixed C, uniformly over integers k with $|k - \log n| \leq C$, the number of isolated vertices is asymptotically Poisson with mean ne^{-k} in total variation and the probability that $LK(n,k)$ is connected is $\exp(-ne^{-k}) + o(1)$. Consequently $LK(n,k)$ is connected with high probability if and only if $k - \log n \to \infty$, and disconnected with high probability if and only if $k - \log n \to -\infty$. The main estimate is that a fixed set of r vertices is isolated with probability $e^{-rk}(1 + o(1))$; it comes from an exact integral formula for the rejections at one vertex and a direct count of the few edges through which the vertices of the set can interact.

1. Introduction

Let $n \geq 3$ and $1 \leq k \leq n-1$. Put independent scores W_e, uniform on $[0,1]$, on the edges e of the complete graph K_n on the vertex set $V = \{1, \dots, n\}$. Each vertex prefers the k edges at it with the smallest scores, and the graph $LK(n,k)$ keeps an edge if and only if both of its endpoints prefer it. The model is due to La and Kabkab [7], who conjectured that for $k = t \log n$ the graph is connected with high probability when $t > 1$ and disconnected when $t < 1$. Moharrami, Subramanian, Liu and Sundaresan [8] identified its local limit for fixed k, the Erlang weighted tree, and studied the corresponding branching process. Dabirian and Subramanian [6] proved the conjecture of La and Kabkab in a sharper form: with high probability,

$$\begin{aligned} &LK(n,k) \text{ is connected} && \text{if } k \geq \log n + t \log\log n\sqrt{\log n} \text{ with } t > \tfrac{1}{2}, \\ &LK(n,k) \text{ is disconnected} && \text{if } k \leq \log n + t \log\log n\sqrt{\log n} \text{ with } t < -\tfrac{1}{2}, \end{aligned}$$

and among their open problems [6, Section 1] they note that it is not known whether the graph is connected when k lies between these two bounds, and suggest that sharper sufficient conditions could close the gap. The answer below has the form of the connectivity threshold of Erdős and Rényi [3], where $G(n, (\log n + c)/n)$ is connected with probability tending to $\exp(-e^{-c})$ and isolated vertices are the last obstruction. Connectivity of the related random k-nearest-neighbour graphs, in which each point of a Poisson process is joined to its k nearest neighbours, is studied by Balister, Bollobás, Sarkar and Walters [1]. In the same model on K_n with independent edge scores, but with an edge kept when either of its endpoints prefers it, connectivity is studied by Cooper and Frieze [2].

Let Z_n be the number of isolated vertices of $LK(n,k)$ and put $\lambda_n = ne^{-k}$.

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Theorem 1.1. *Fix $C < \infty$. As $n \to \infty$, uniformly over integers k with $|k - \log n| \le C$,*

$$d_{\mathrm{TV}}(\mathcal{L}(Z_n), \mathrm{Poi}(\lambda_n)) \to 0 \qquad \text{and} \qquad \mathbb{P}\big(LK(n,k) \text{ is connected}\big) - e^{-\lambda_n} \to 0.$$

Corollary 1.2. *Let $k = k(n)$. If $k - \log n \to \infty$, then $LK(n,k)$ is connected with probability tending to one. If $k - \log n \to -\infty$, then $LK(n,k)$ is disconnected with probability tending to one. If $k - \log n \to c \in \mathbb{R}$ along a subsequence, then along it Z_n converges in distribution to $\mathrm{Poi}(e^{-c})$ and the probability of connectivity tends to $\exp(-e^{-c})$.*

So the threshold is at $\log n$ with an additive window of width $O(1)$, and inside it the only obstruction to connectivity is an isolated vertex, as in the Erdős–Rényi graph. The scale e^{-k} is easy to see. A vertex v is isolated when each of its k preferred edges is refused at the other end. If the edge to u has score x, then u refuses it when u has at least k other edges lighter than x, which in the limit has the probability $F_k(s) = \mathbb{P}(\Gamma_k \le s)$, where $s = nx$ and Γ_k has the gamma law of shape k. The scaled scores at v are the points $s_1 < s_2 < \cdots$ of a Poisson process of rate one, and Lemma 2.1 below turns $\mathbb{E}\prod_{j \le k} F_k(s_j)$ into

$$\frac{1}{k!} \int_0^\infty \Big(\int_0^s F_k \Big)^k e^{-s}\, ds.$$

For large s the inner integral is $\mathbb{E}(s - \Gamma_k)_+ \approx s - k$, and $\int_k^\infty (s-k)^k e^{-s}\, ds/k! = e^{-k}$ exactly. The integrand peaks at $s \approx 2k$, where $s - k$ is a good approximation, so the integral is $e^{-k}(1 + o(1))$. It is the limit of the quantity M of Lemma 2.2, in which the refusals at the other ends are treated as independent. The quantity M is not the probability that v is isolated: for $n = 4$ and $k = 2$, with $m = 3$ and $N = 2$, $M = 1/21$, while v is isolated with probability $1/15$. Proposition 3.1 shows that the two agree to first order, so that the isolation probability is also $e^{-k}(1 + o(1))$. At $k = \log n$ this is of order $1/n$. Every event through which several vertices interact (two of them preferring the same outside vertex, a light edge between two of the outside vertices involved) costs a further factor $(\log n)^{O(1)}/n$. Only boundedly many of them can occur at a cost comparable to e^{-rk}, and each removes at most a bounded number of the rk refusals that isolation requires; Lemma 2.3 shows that removing d refusals multiplies the probability by at most $(3k)^d$. This is the content of Proposition 3.1. Components with between 2 and 6 vertices are handled by the same proposition, larger components by a first-moment bound.

Section 2 treats one vertex, Section 3 several vertices and the Poisson limit, Section 4 the components, and Section 5 proves Theorem 1.1 and Corollary 1.2. Section 6 describes the computer checks that accompany the paper.

Throughout, C is fixed, $n \to \infty$, and k is an integer with $|k - \log n| \le C$; in particular $k \to \infty$ and $k^a/n \to 0$ for every a. All $o(1)$ terms are uniform in such k and in the other parameters as stated. We write $(m)_j = m(m-1)\cdots(m-j+1)$ and use the Chernoff bounds for a binomial variable X with mean λ,

$$\mathbb{P}(X \ge t) \le e^{-\lambda\varphi(t/\lambda)}\ (t \ge \lambda), \qquad \mathbb{P}(X \le t) \le e^{-\lambda\varphi(t/\lambda)}\ (t \le \lambda), \qquad \varphi(y) = y\log y - y + 1. \tag{1.1}$$

The jth smallest of N independent uniform variables has the $\mathrm{Beta}(j, N-j+1)$ law, and it is at most x exactly when $\mathrm{Bin}(N,x) \ge j$.

2. One vertex

Lemma 2.1. *Let $m > k \ge 1$, let $X_1 < \cdots < X_k$ be the k smallest of m independent uniform variables on $[0,1]$, let $g\colon [0,1] \to [0,1]$ be nondecreasing and put $G(x) = \int_0^x g$. Then*

$$\mathbb{E}\prod_{i=1}^k g(X_i) = \frac{(m)_{k+1}}{k!} \int_0^1 G(x)^k (1-x)^{m-k-1}\, dx. \tag{2.1}$$

If $0 \le d \le k-1$ and $A \subseteq \{1,\dots,k\}$ has $k-d$ elements, then

$$\mathbb{E}\prod_{i\in A} g(X_i) \le \binom{k-1}{d}\frac{(m)_k}{(k-1)!}\int_0^1 g(x)G(x)^{k-1-d}x^d(1-x)^{m-k}\,dx. \tag{2.2}$$

Proof. Let $a_1 < \cdots < a_{k-d}$ be the elements of A. At least $k-d-i$ elements of A exceed a_i, so $a_i \le i+d$, and since g is nondecreasing, $\prod_{i\in A} g(X_i) \le \prod_{i=d+1}^{k} g(X_i)$. The variable X_k has density $\frac{(m)_k}{(k-1)!}x^{k-1}(1-x)^{m-k}$, and given $X_k = x$ the set $\{X_1,\dots,X_{k-1}\}$ consists of $k-1$ independent points $Y_1,\dots,Y_{k-1}$, uniform on $[0,x]$. The product $\prod_{i=d+1}^{k-1} g(X_i)$ is one of the $\binom{k-1}{d}$ products of $g(Y_b)$ over sets of $k-1-d$ indices b, and each such product has conditional mean $(G(x)/x)^{k-1-d}$. Hence

$$\mathbb{E}\Big[\prod_{i=d+1}^{k} g(X_i)\;\Big|\; X_k = x\Big] \le \binom{k-1}{d} g(x)\Big(\frac{G(x)}{x}\Big)^{k-1-d},$$

with equality when $d = 0$. Integrating against the density of X_k gives (2.2). For $d = 0$ this is an equality, and since $gG^{k-1} = (G^k)'/k$, $G(0) = 0$ and $m > k$, an integration by parts gives (2.1). □

Fix $A_0 < \infty$. In the rest of this section m and N are integers with $|m-n| \le A_0 k$ and $|N-n| \le A_0 k$, and

$$q(x) = \mathbb{P}(\mathrm{Bin}(N,x) \ge k) = \mathbb{P}(B \le x), \quad Q(x) = \int_0^x q = \mathbb{E}(x-B)_+, \quad B \sim \mathrm{Beta}(k, N-k+1).$$

The function q is nondecreasing with values in $[0,1]$, and Q is convex with $Q(0) = 0$, so $Q(x)/x$ is nondecreasing. Put

$$\mu = \mathbb{E}B = \frac{k}{N+1}, \qquad \sigma^2 = \mathrm{Var}\,B = \frac{k(N+1-k)}{(N+1)^2(N+2)} \le \frac{\mu^2}{k},$$

and let $X_1 < \cdots < X_k$ be the k smallest of m independent uniform variables.

Lemma 2.2. $M := \mathbb{E}\prod_{i=1}^{k} q(X_i) = e^{-k}(1+o(1))$.

Proof. By Jensen's inequality $Q(x) \ge (x-\mu)_+$. With the substitution $x = \mu + (1-\mu)z$,

$$\frac{(m)_{k+1}}{k!}\int_\mu^1 (x-\mu)^k(1-x)^{m-k-1}\,dx = (1-\mu)^m\frac{(m)_{k+1}}{k!}\int_0^1 z^k(1-z)^{m-k-1}\,dz = (1-\mu)^m. \tag{2.3}$$

Since $m\mu = k\,m/(N+1) = k + O(k^2/n)$ and $m\mu^2 = O(k^2/n)$, we have $(1-\mu)^m = e^{-k}(1+o(1))$, and (2.1) gives $M \ge e^{-k}(1+o(1))$.

For the upper bound split the integral in (2.1) at μ and at $\mu + a_1$, where $a_1 = \mu k^{-1/3}$. We use $m\mu \le 2k$, $m\sigma \le m\mu/\sqrt{k} \le 2\sqrt{k}$ and $k! \ge (k/e)^k$. For $x \le \mu$ we have $Q(x) \le Q(\mu) = \mathbb{E}(\mu - B)_+ \le \sigma$, so this part is at most

$$\frac{m^{k+1}}{k!}\mu\sigma^k \le 2k\frac{(2\sqrt{k})^k}{k!} \le 2k\Big(\frac{2e}{\sqrt{k}}\Big)^k = o(e^{-k}).$$

For $x \ge \mu$ write $a = x - \mu$ and $R(x) = \mathbb{E}(B-x)_+$, so that $Q(x) = a + R(x)$ and $R(x) \le R(\mu) \le \sigma \le \mu k^{-1/2}$. For $a \le a_1$ this gives $Q(x) \le 2\mu k^{-1/3}$, and that part of the integral is at most

$$\frac{m^{k+1}}{k!}a_1(2\mu k^{-1/3})^k \le 2k\frac{(4k)^k k^{-k/3}}{k!} \le 2k\,(4e)^k k^{-k/3} = o(e^{-k}).$$

On $a > a_1$ write $Q^k = a^k(1+\rho)^k$ with $\rho = R/a$. By (2.3), the remaining part equals $(1-\mu)^m \mathbb{E}[(1+\rho)^k;\, a > a_1]$, where now $a = (1-\mu)\zeta$ with $\zeta \sim \mathrm{Beta}(k+1, m-k)$. Two estimates for ρ cover this range.

If $a_1 < a < \mu/2$, use $(y-a)_+ \le y^2/(4a)$, which holds for all real y because $y^2 - 4a(y-a) = (y-2a)^2$. With $y = B - \mu$ it gives $R(x) \le \sigma^2/(4a)$, so $\rho \le \sigma^2/(4a_1^2) \le k^{-1/3}/4$ and $(1+\rho)^k \le e^{k^{2/3}/4}$. On the other hand $\mathbb{P}(\zeta \le y) = \mathbb{P}(\mathrm{Bin}(m, y) \ge k+1)$ with $y = \mu/(2(1-\mu))$, and $my = k/2 + O(k^2/n)$, so by (1.1) this probability is at most $\exp(-(k/2)\varphi(2)(1+o(1)))$, where $\varphi(2) = 2\log 2 - 1 > 0$. The contribution of this range to $\mathbb{E}(1+\rho)^k$ is therefore at most $\exp(k^{2/3}/4 - k\varphi(2)/3) = o(1)$.

If $a \ge \mu/2$, then $x \ge 3\mu/2$. Since $x\, f_{k,N-k+1}(x) = \mu f_{k+1,N-k+1}(x)$ for the beta densities $f_{\alpha,\beta}$, we have $R(x) \le \mathbb{E}[B; B > x] = \mu \mathbb{P}(B^+ > x)$ with $B^+ \sim \mathrm{Beta}(k+1, N-k+1)$. Now $\mathbb{P}(B^+ > 3\mu/2) = \mathbb{P}(\mathrm{Bin}(N+1, 3\mu/2) \le k)$, a binomial variable of mean $3k/2$, so by (1.1) it is at most $e^{-\gamma k}$ with $\gamma = \frac{3}{2}\varphi(\frac{2}{3}) = \log\frac{2}{3} + \frac{1}{2} > 0$. Hence $\rho \le 2e^{-\gamma k}$ and $(1+\rho)^k \le \exp(2ke^{-\gamma k}) = 1 + o(1)$.

Together, $\mathbb{E}[(1+\rho)^k; a > a_1] \le 1 + o(1)$, and $M \le (1-\mu)^m(1+o(1)) + o(e^{-k}) = e^{-k}(1+o(1))$. □

The value $\gamma = \log(2/3) + 1/2 \approx 0.095$ is small, but only $\gamma > 0$ is used. The next lemma removes refusals at arbitrary ranks.

Lemma 2.3. *Fix $D < \infty$. Uniformly over $0 \le d \le D$ and over sets $A \subseteq \{1, \dots, k\}$ with $k - d$ elements,*

$$\mathbb{E} \prod_{i \in A} q(X_i) \le (3k)^d e^{-k}(1 + o(1)).$$

Proof. Let $J_d = \frac{(m)_k}{(k-1)!} \int_0^1 q Q^{k-1-d} x^d (1-x)^{m-k}\, dx$, so that $J_0 = M$ by Lemma 2.1 and the left side is at most $\binom{k-1}{d} J_d \le k^d J_d$. For $x \ge 3\mu/2$, $Q(x) \ge x - \mu \ge x/3$, so $qQ^{k-1-d}x^d \le 3^d q Q^{k-1}$. For $x < 3\mu/2$, the monotonicity of $Q(x)/x$ gives

$$\frac{Q(x)}{x} \le \frac{Q(3\mu/2)}{3\mu/2} \le \frac{\mu/2 + \mathbb{E}(\mu - B)_+}{3\mu/2} \le \frac{1}{3} + \frac{2\sigma}{3\mu} \le \frac{1}{3}\Big(1 + \frac{2}{\sqrt{k}}\Big) =: \theta,$$

using $(3\mu/2 - B)_+ \le \mu/2 + (\mu - B)_+$. There $qQ^{k-1-d}x^d \le x^{k-1}\theta^{k-1-d}$, and $\frac{(m)_k}{(k-1)!} x^{k-1}(1-x)^{m-k}$ is a probability density. Therefore

$$J_d \le 3^d M + \theta^{k-1-d} \le 3^d M + 3^{D+1} e^{-k\log 3 + 2\sqrt{k}}.$$

Since $\log 3 > 1$, the last term is $o(e^{-k})$, and Lemma 2.2 completes the proof. □

3. Several vertices

Fix $r \ge 1$ and a set $R = \{v_1, \dots, v_r\}$ of vertices; let $V' = V \setminus R$. For each $a \le r$ let $X_{a,1} < X_{a,2} < \cdots < X_{a,n-r}$ be the scores of the $n - r$ edges from v_a to V', and let $U_{a,j} \in V'$ be the other endpoint of the edge with score $X_{a,j}$. Put $T_a = X_{a,k}$ and $T = \max_a T_a$. Call the data $(X_{a,j}, U_{a,j})_{a \le r,\, j \le k}$ the *stars*. The following facts are standard.

(F1) The r families $(X_{a,j}, U_{a,j})_{j \le k}$ are independent. In each, $(X_{a,j})_{j \le k}$ are the k smallest of $n - r$ independent uniform variables, and $(U_{a,j})_{j \le k}$ is a uniformly random sequence of k distinct elements of V', independent of the scores.

(F2) Given the stars, the remaining edge scores are independent; the score of an edge $\{v_a, w\}$ with $w \in V' \setminus \{U_{a,1}, \dots, U_{a,k}\}$ is uniform on $[T_a, 1]$, and every other remaining edge has a uniform score on $[0, 1]$.

For $1 \le p \le k$ let $\mathcal{A}_R(p)$ be the event that for every $a \le r$ and $j \le p$ the vertex $u = U_{a,j}$ has at least k edges other than $\{u, v_a\}$ with score less than $X_{a,j}$; that is, u refuses the jth lightest edge from v_a to V'. This event does not involve the edges between vertices of R.

Proposition 3.1. *Fix $r \geq 1$ and $e \geq 0$. Uniformly over sets R of size r,*

$$\mathbb{P}(\mathcal{A}_R(k)) = e^{-rk}(1+o(1)) \qquad \text{and} \qquad \mathbb{P}(\mathcal{A}_R(k-e)) = O(k^{re}e^{-rk}).$$

Proof. Let $S = \{U_{a,j} : a \leq r,\ j \leq k\}$, let $c = rk - |S|$ be the number of coincidences among the labels, and let $W = V' \setminus S$. For each (a,j) with $j \leq k$ let $N_{a,j}$ be the number of $w \in W$ with $W_{\{U_{a,j},w\}} < X_{a,j}$. By (F2), given the stars, the edges between S and W have independent uniform scores, and edges at distinct vertices of S are distinct.

Lower bound. The vertex $U_{a,j}$ has at least $N_{a,j}$ edges other than $\{U_{a,j}, v_a\}$ below $X_{a,j}$, because $v_a \notin W$. On the event $c = 0$ the variables $N_{a,j}$ are, given the stars, independent with laws $\mathrm{Bin}(n-r-rk, X_{a,j})$. Hence, with $q_0(x) = \mathbb{P}(\mathrm{Bin}(n-r-rk, x) \geq k)$ and (F1),

$$\mathbb{P}(\mathcal{A}_R(k)) \geq \mathbb{E}\Big[\mathbf{1}\{c=0\}\prod_{a,j} q_0(X_{a,j})\Big] = \mathbb{P}(c=0)\, M_0^r,$$

where $M_0 = \mathbb{E}\prod_{j\leq k} q_0(X_{1,j}) = e^{-k}(1+o(1))$ by Lemma 2.2 with $m = n-r$ and $N = n-r-rk$. Two given labels of different vertices of R coincide with probability $1/(n-r)$, so $\mathbb{P}(c=0) \geq 1 - r^2k^2/(n-r) = 1-o(1)$.

Upper bound. Fix $B = 2r+6$, so that $B - 1 - \log B \geq B/2 > r+1$, and put $\tau = Bk/n$. By (1.1), $\mathbb{P}(T_a > \tau) = \mathbb{P}(\mathrm{Bin}(n-r,\tau) \leq k-1) \leq e^{-k(B-1-\log B)(1+o(1))} = o(e^{-rk})$, so we may work on $\{T \leq \tau\}$.

Let I be the set of edges with both endpoints in $R \cup S$, other than the edges between two vertices of R and other than the star edges $\{v_a, U_{a,j}\}$, $j \leq k$. Call an edge of I *light* if its score is at most τ, and let Λ be the number of light edges. Since $|I| \leq (r+rk)^2 \leq 4r^2k^2$ and, by (F2), on $\{T \leq \tau\}$ each edge of I is light with conditional probability at most $\tau/(1-\tau) \leq 2\tau$, independently,

$$\mathbb{P}(\Lambda \geq \ell \mid \text{stars}) \leq (8r^2k^2\tau)^\ell = (8Br^2k^3/n)^\ell \qquad \text{on } \{T \leq \tau\}. \tag{3.1}$$

By (F1), revealing the labels in order, each label repeats one of the at most rk earlier labels of other vertices of R with conditional probability at most $rk/(n-r-k)$, so

$$\mathbb{P}(c \geq \gamma) \leq \binom{rk}{\gamma}\Big(\frac{rk}{n-r-k}\Big)^\gamma \leq \Big(\frac{2r^2k^2}{n}\Big)^\gamma. \tag{3.2}$$

Taking $\ell = r+1$ and $\gamma = r+1$ shows that $\mathbb{P}(\Lambda \geq r+1,\, T \leq \tau) + \mathbb{P}(c \geq r+1) = O(k^{3r+3}n^{-r-1}) = o(e^{-rk})$, because $e^{-rk} \geq e^{-rC}n^{-r}$.

Call an occurrence (a,j), $j \leq k$, *clean* if its label $u = U_{a,j}$ is the label of no other occurrence and u is not an endpoint of a light edge. At most $2c$ occurrences have a repeated label, since a label carried by $h \geq 2$ occurrences accounts for $h-1$ coincidences and $h \leq 2(h-1)$. Among the occurrences with unrepeated labels, each endpoint of a light edge is the label of at most one, so at most 2Λ of them have a label on a light edge. Hence at most $2c + 2\Lambda$ occurrences are not clean. If (a,j) is clean, then every edge from u to $(R \cup S) \setminus \{u, v_a\}$ lies in I and is not light: an edge $\{u, v_b\}$ with $b \neq a$ is not a star edge because u is not a label of v_b. On $\{T \leq \tau\}$ all these edges have scores above $\tau \geq X_{a,j}$. Therefore on $\mathcal{A}_R(p) \cap \{T \leq \tau\}$ every clean occurrence (a,j) with $j \leq p$ has $N_{a,j} \geq k$.

Fix $\ell, \gamma \leq r$. Given the stars, the light edges are determined by the scores on I, which are independent of the scores between S and W; the clean occurrences carry distinct labels; and $N_{a,j}$ is stochastically dominated by $\mathrm{Bin}(n-r, X_{a,j})$. Hence, with $q_1(x) = \mathbb{P}(\mathrm{Bin}(n-r,x) \geq k)$, on $\{T \leq \tau,\, c = \gamma\}$,

$$\mathbb{P}(\mathcal{A}_R(p),\, \Lambda = \ell \mid \text{stars}) \leq \mathbb{P}(\Lambda \geq \ell \mid \text{stars}) \max_F \prod_{(a,j)\notin F,\ j\leq p,\ (a,j)\text{ unrepeated}} q_1(X_{a,j}),$$

the maximum being over sets F of at most 2ℓ occurrences. At vertex a the product runs over at least $k-e_a$ ranks, where $e_a=(k-p)+g_a+f_a$, g_a is the number of repeated occurrences of a and f_a the number of occurrences of a in F; since $q_1\le 1$, the first part of the proof of Lemma 2.1 bounds it by $\prod_{i>e_a} q_1(X_{a,i})$. The vector (f_a) takes at most $(2\ell+1)^r$ values. The labels are independent of the scores, so by (3.1), (3.2), (F1) and Lemma 2.3 with $m=N=n-r$ and $D=re+4r$,

$$\mathbb{P}(\mathcal{A}_R(k-e),\,T\le\tau,\,\Lambda=\ell,\,c=\gamma)\le\Big(\frac{8Br^2k^3}{n}\Big)^{\ell}\Big(\frac{2r^2k^2}{n}\Big)^{\gamma}(2\ell+1)^r(3k)^{re+2\gamma+2\ell}e^{-rk}(1+o(1)).$$

For $\ell+\gamma\ge 1$ this is $O(k^{re}e^{-rk}(\log n)^{O(1)}/n)$. For $\ell=\gamma=0$ every occurrence with $j\le k-e$ is clean, and the bound is $(3k)^{re}e^{-rk}(1+o(1))$; for $e=0$ it is exactly $\mathbb{E}\prod_{a,j}q_1(X_{a,j})=M_1^r$, where $M_1=e^{-k}(1+o(1))$ by Lemma 2.2. Summing over $\ell,\gamma\le r$ and adding the bounds for $T>\tau$, $\Lambda\ge r+1$ and $c\ge r+1$ proves both assertions. □

Corollary 3.2. *Fix $r\ge 1$. Uniformly over distinct vertices $v_1,\dots,v_r$, the probability that all of them are isolated in $LK(n,k)$ is $e^{-rk}(1+o(1))$.*

Proof. Let $\mathcal{I}$ be the event that $v_1,\dots,v_r$ are isolated, and let $\mathcal{D}$ be the event that for every a every edge from v_a to another vertex of R has score greater than T_a. On $\mathcal{D}$ the k edges preferred by v_a are its k lightest edges to V', so $\mathcal{I}\cap\mathcal{D}=\mathcal{A}_R(k)\cap\mathcal{D}$. In general v_a prefers at most $r-1$ edges inside R, so it prefers its $k-r+1$ lightest edges to V', and $\mathcal{I}\subseteq\mathcal{A}_R(k-r+1)$. The event $\mathcal{D}$ depends on the edges inside R and on the stars, and $\mathcal{A}_R(p)$ does not involve the edges inside R; by (F2) the two are independent given the stars, and $\mathbb{P}(\mathcal{D}^c\mid\text{stars})\le r^2T$. With τ as in the proof of Proposition 3.1,

$$\mathbb{P}(\mathcal{I})\ge\mathbb{P}(\mathcal{A}_R(k)\cap\mathcal{D})\ge(1-r^2\tau)\big(\mathbb{P}(\mathcal{A}_R(k))-\mathbb{P}(T>\tau)\big)=e^{-rk}(1+o(1)),$$

$$\mathbb{P}(\mathcal{I})\le\mathbb{P}(\mathcal{A}_R(k))+r^2\tau\,\mathbb{P}(\mathcal{A}_R(k-r+1))+\mathbb{P}(T>\tau)=e^{-rk}(1+o(1))+O\Big(\frac{k^{r^2+1}}{n}\Big)e^{-rk}.\quad\square$$

Proposition 3.3. *Uniformly over $|k-\log n|\le C$, $d_{\mathrm{TV}}(\mathcal{L}(Z_n),\mathrm{Poi}(\lambda_n))\to 0$.*

Proof. By Corollary 3.2, for each fixed r the factorial moment $\mathbb{E}(Z_n)_r=(n)_r\mathbb{P}(v_1,\dots,v_r\text{ isolated})$ equals $\lambda_n^r(1+o(1))$, and $\lambda_n\in[e^{-C},e^{C}]$. Suppose the claim fails. Then there are $\varepsilon>0$ and a sequence $n_i\to\infty$ with admissible k_i along which the distance stays above ε; passing to a subsequence, $\lambda_{n_i}\to\lambda\in[e^{-C},e^{C}]$. All factorial moments converge to those of $\mathrm{Poi}(\lambda)$, which is determined by its moments, so Z_{n_i} converges in distribution to $\mathrm{Poi}(\lambda)$ [4, Theorem 30.2], hence in total variation since the variables are integer valued. As $d_{\mathrm{TV}}(\mathrm{Poi}(\lambda_{n_i}),\mathrm{Poi}(\lambda))\le|\lambda_{n_i}-\lambda|\to 0$, this is a contradiction. □

4. Components with at least two vertices

Let τ_v be the kth smallest score at the vertex v, and put $a=5/32$, $b=32/5$. Let $\mathcal{G}$ be the event that $ak/(n-1)\le\tau_v\le bk/(n-1)$ for every v. By (1.1),

$$\mathbb{P}(\tau_v<ak/(n-1))\le e^{-k(a-1-\log a)},\qquad\mathbb{P}(\tau_v>bk/(n-1))\le e^{-k(b-1-\log b)},$$

and $a-1-\log a=1.0125\ldots$, $b-1-\log b=3.543\ldots$. A union bound over the n vertices gives $\mathbb{P}(\mathcal{G}^c)=O(n^{-0.0125})$.

On $\mathcal{G}$ an edge with score below $ak/(n-1)$ is preferred at both ends, hence kept, and a kept edge has score at most $bk/(n-1)$. So if a set S of s vertices is the vertex set of a component and $\mathcal{G}$ holds, then every edge between S and $V\setminus S$ has score at least $ak/(n-1)$, and the kept edges inside S contain a spanning tree whose edges have scores at most $bk/(n-1)$.

Lemma 4.1. *The probability that $\mathcal{G}$ holds and $LK(n,k)$ has a component with s vertices for some $7 \le s \le n/2$ tends to zero.*

Proof. For a fixed set S of size s the two events above involve disjoint sets of edges. There are s^{s-2} spanning trees on S, so

$$\mathbb{P}(S \text{ is a component, } \mathcal{G}) \le s^{s-2}\Big(\frac{bk}{n-1}\Big)^{s-1}\exp\Big(-\frac{aks(n-s)}{n-1}\Big). \tag{4.1}$$

For $7 \le s \le 12$ the exponential factor is $O(n^{-as})$, and summing over the $\binom{n}{s} \le n^s$ sets gives $O(n^{1-as}k^{s-1}) = O(n^{-3/32}k^{11})$, since $7a = 35/32$. For $13 \le s \le n/2$ we use $\binom{n}{s}s^{s-2} \le (en)^s$, $(bk/(n-1))^{s-1} \le (2bk/n)^{s-1}$ and $(n-s)/(n-1) \ge 1/2$, so that the exponential factor is at most $(e^{aC/2}n^{-a/2})^s$. With $y_n = 2ebke^{aC/2}n^{-a/2} \to 0$ the sum over these s is at most

$$\frac{n}{2bk}\sum_{s\ge 13} y_n^s \le \frac{n\,y_n^{13}}{2bk(1-y_n)} = O(n^{1-13a/2}k^{12}) = O(n^{-1/64}k^{12}). \qquad \square$$

Lemma 4.2. *For each $2 \le s \le 6$, the probability that $\mathcal{G}$ holds and $LK(n,k)$ has a component with exactly s vertices tends to zero.*

Proof. Fix a set S of s vertices and use the notation of Section 3 with $R = S$. A vertex $v \in S$ has $s-1$ edges inside S, so among the edges it prefers are its $k-s+1$ lightest edges to $V \setminus S$. If S is the vertex set of a component, none of these edges is kept, so each is refused at its other end, and $\mathcal{A}_S(k-s+1)$ holds. That event involves no edge inside S, while on $\mathcal{G}$ the tree event of the previous lemma involves only edges inside S. By Proposition 3.1 with $r = s$ and $e = s-1$,

$$\mathbb{P}(S \text{ is a component, } \mathcal{G}) \le s^{s-2}\Big(\frac{bk}{n-1}\Big)^{s-1}O(k^{s(s-1)}e^{-sk}) = O\big(k^{s^2}n^{-2s+1}\big),$$

using $e^{-k} \le e^C/n$. Summing over $\binom{n}{s} \le n^s$ sets gives $O(k^{s^2}n^{1-s}) \to 0$. $\square$

5. Proofs of Theorem 1.1 and Corollary 1.2

Proof of Theorem 1.1. The first assertion is Proposition 3.3, which also gives $\mathbb{P}(Z_n = 0) = e^{-\lambda_n} + o(1)$. A disconnected graph without isolated vertices has a component with between 2 and $n/2$ vertices, so

$$0 \le \mathbb{P}(Z_n = 0) - \mathbb{P}(LK(n,k) \text{ connected}) \le \mathbb{P}(\mathcal{G}^c) + \sum_{s=2}^{\lfloor n/2\rfloor}\mathbb{P}(\mathcal{G}, \text{ a component with } s \text{ vertices}),$$

which tends to zero uniformly by Lemmas 4.1 and 4.2. $\square$

Proof of Corollary 1.2. Use one array of scores for all k. An edge preferred at both ends for the budget k is still preferred for the budget $k+1$, so $LK(n,k) \subseteq LK(n,k+1)$ and connectivity is increasing in k. If $k - \log n \to \infty$, fix $L > 0$ and let q_n be the integer in $[\log n + L, \log n + L + 1)$; eventually $q_n \le k$, and Theorem 1.1 with $C = L + 1$ gives

$$\liminf_n \mathbb{P}(LK(n,k) \text{ connected}) \ge \liminf_n \mathbb{P}(LK(n,q_n) \text{ connected}) \ge \exp(-e^{-L}),$$

which tends to one as $L \to \infty$. If $k - \log n \to -\infty$, let q_n be the integer in $(\log n - L - 1, \log n - L]$; eventually $q_n \ge k$, and the limsup of the probability of connectivity is at most $\exp(-e^L) \to 0$. The last assertion is Theorem 1.1 with $C = |c| + 1$, since then $\lambda_n \to e^{-c}$. If $k - \log n$ does not tend to ∞, some subsequence has $k - \log n \to -\infty$ or $k - \log n \to c \in \mathbb{R}$, and along it the probability of connectivity tends to 0 or to $\exp(-e^{-c}) < 1$; so connectivity with high probability holds exactly when $k - \log n \to \infty$, and in the same way disconnection with high probability holds exactly when $k - \log n \to -\infty$. $\square$

Since $\log n \pm t \log\log n\sqrt{\log n} - \log n \to \pm\infty$ for every $t > 0$, the corollary settles every sequence $k(n)$ in the window of [6].

6. Computations

The repository [5] contains two Python programs. The first, `checks.py`, verifies the numerical constants of the paper in exact rational arithmetic ($a - 1 - \log a > 1$ for $a = 5/32$, $b - 1 - \log b > 1$ for $b = 32/5$, $B - 1 - \log B \geq B/2$ for $B \geq 6$, $\log 3 > 1$, $\varphi(2) > 0$ and $\varphi(2/3) > 0$, $7a > 1$, $13a/2 > 1$), tests (2.1) and (2.2) against Monte Carlo, and evaluates $M\, e^k$ of Lemma 2.2 by numerical integration for n up to 10^7.

The second program, `simulate.py`, first evaluates the Poisson-process form of M described in the introduction, $I_k = \frac{1}{k!}\int_0^\infty (\mathbb{E}(s - \Gamma_k)_+)^k e^{-s}\, ds$; integrating e^k times the integrand with absolute and relative tolerances 10^{-13} and 10^{-12}, it gives $I_k e^k = 1.02974$, 1.00240 and 1.00022 at $k = 10$, 20 and 30. It then samples $LK(n, k)$ directly. Table 1 compares the empirical probability of connectivity with $e^{-\lambda_n}$. In every sample with $n = 3000$, and in every sample with $n = 1000$ and $k \geq 7$, a graph with no isolated vertex was connected.

n	k	samples	$\lambda_n = ne^{-k}$	$\mathbb{P}$(connected)	$e^{-\lambda_n}$
1000	6	300	2.479	0.067 ± 0.014	0.084
1000	7	300	0.912	0.413 ± 0.028	0.402
1000	8	300	0.335	0.713 ± 0.026	0.715
1000	9	300	0.123	0.910 ± 0.017	0.884
3000	7	200	2.736	0.035 ± 0.013	0.065
3000	8	200	1.006	0.395 ± 0.035	0.366
3000	9	200	0.370	0.705 ± 0.032	0.691
3000	10	200	0.136	0.880 ± 0.023	0.873

Table 1. Simulated probability of connectivity of $LK(n, k)$, with one standard error.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [9] and working together inside the Brain framework [11], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [10], the Math Brain repository [9] and the Brain framework repository [11].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In September 2026 OpenAI GPT models, running as research tasks of that project, proposed Theorem 1.1, the integral formula (2.1), Lemma 2.2, the treatment of components in Section 4, and a proof of the multi-vertex estimate through a canonical deletion of interacting edges. In October 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, checked that proof, replaced its multi-vertex part by the argument of Lemma 2.3 and Proposition 3.1, wrote the computer checks of the repository [5], and wrote the paper. In November 2026 a Codex agent (OpenAI, GPT-6.1) refereed the manuscript, checking every lemma and the numerical claims with its own programs. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code is available at `https://github.com/yspennstate/bilateral-preference-connectivity-window`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [9].

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Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel

Email address: `yitzchak.shmalo@gmail.com`